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Course: CSE-440

Section: 01

Project group: 07

Poker Game Simulator using Expectiminimax

Week 1 (10 Feb – 16 Feb) I set up the development environment and configured GitHub access so I could pull and run the project code on my own machine. I reviewed the course manual to understand repository requirements (main file, folders, README, etc.). I studied basic concepts of playing cards, shuffling, and the idea of Expectiminimax so I could support the implementation work.

Week 2 (17 Feb – 23 Feb) I helped refine the data structures for representing cards, players, and the overall game state, focusing on clarity and avoiding redundancy. I implemented small utility functions such as converting internal card codes into readable strings for printing, resetting game variables between rounds, and simple helper checks used across the project. I tested these utilities individually to confirm that they behave correctly in isolation.

Week 3 (24 Feb – 1 Mar) When the first version of the game engine was available, I mainly concentrated on debugging support. I repeatedly ran the program with different random seeds to look for crashes or inconsistent states. I added temporary debug prints that show current player, pot size, and cards at each step. Using these logs, I reported several issues regarding turn order and incorrect pot updates, which the leader then fixed.

Week 4 (2 Mar – 8 Mar) I collaborated with the leader on designing the numeric scoring scheme used for hand evaluation. I helped decide ranges that distinguish weak, medium, and strong hands while keeping the values suitable for tree-search utilities. I checked the scoring function on many random samples and confirmed that stronger hands consistently receive higher scores and that differences between close hands are not lost.

Week 5 (9 Mar – 15 Mar) After the Expectiminimax AI was connected, I organized informal matches where the bot played against simple scripted behaviors. I collected statistics such as win/loss counts and typical betting patterns and identified some situations where the bot's decisions seemed unintuitive. I summarized these observations so that we can adjust parameters like search depth and evaluation scaling in future iterations.

Current Status and Next Steps

So far, I have contributed helper utilities, debugging support, scoring-scheme design, and empirical testing of the integrated AI. Next, I plan to assist in parameter tuning, help prepare tables and figures for the reports, and ensure that our repository meets the required structure